



Cyber

Protect

Data

Threat

Security

Attack

Firewall

Malware

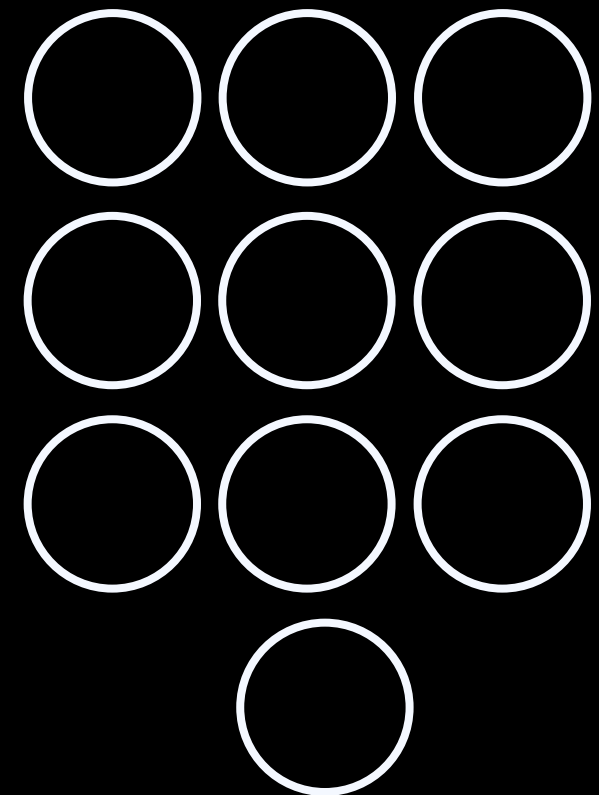
THE SOLARWINDS CYBER ATTACK

Cyber Sphere and Security

by Sachin Nawadia



B.com(H) 3558





SOLARWINDS



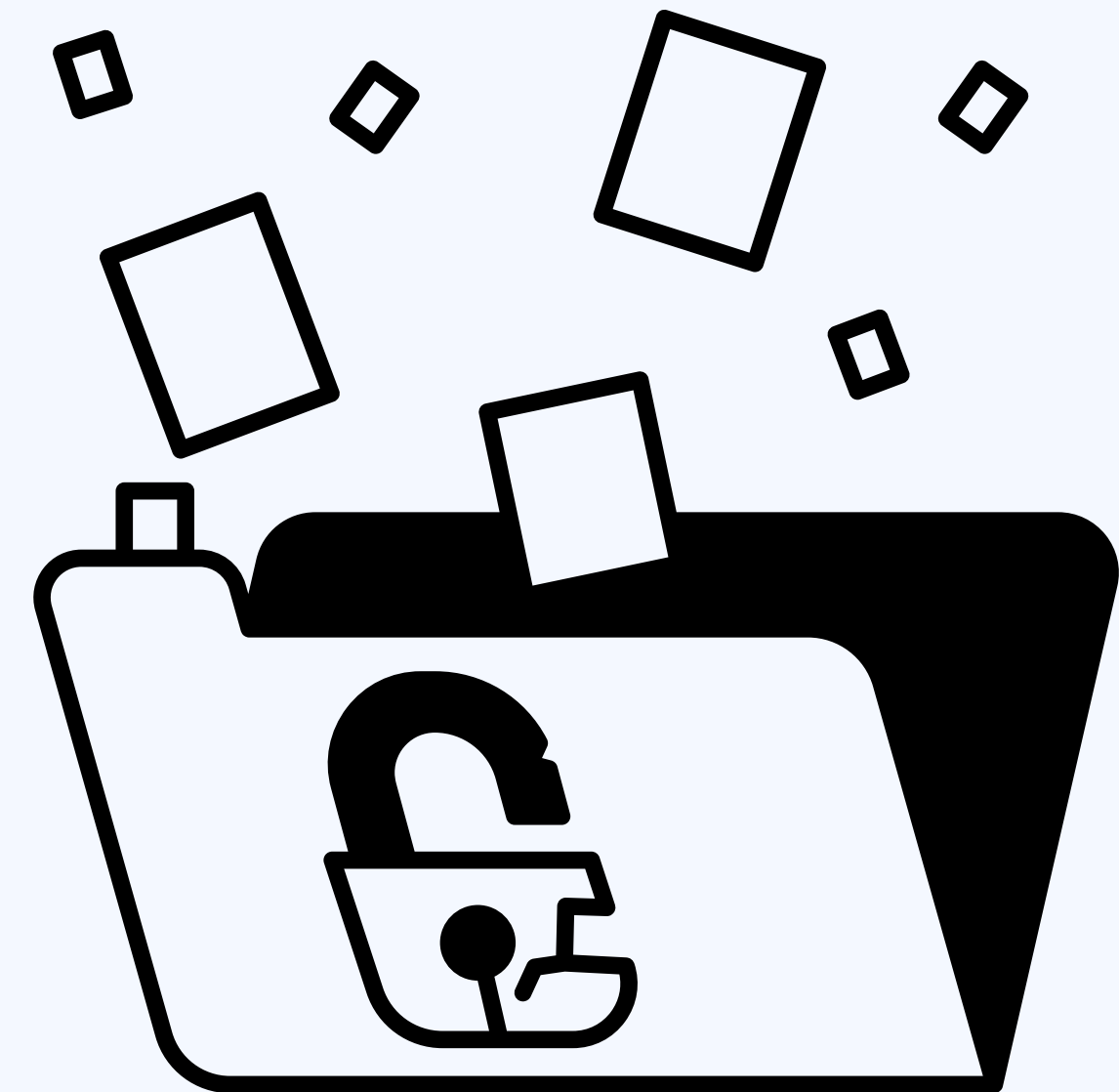
SolarWinds is a global IT management software company, headquartered in Austin, Texas, that provides solutions for observability, database management, service management, and IT security, serving businesses of all sizes.



HOW IT WORKED

How Cyber Attacks Happen

Hackers gained access to SolarWinds' systems in late 2019 and deployed a backdoor, known as "Sunburst," into the Orion software updates. When SolarWinds customers installed these compromised updates, the malicious code was deployed on their networks, giving the attackers access to their systems.



TIMELINE

1

Hackers gained access to SolarWinds' systems in late 2019

2

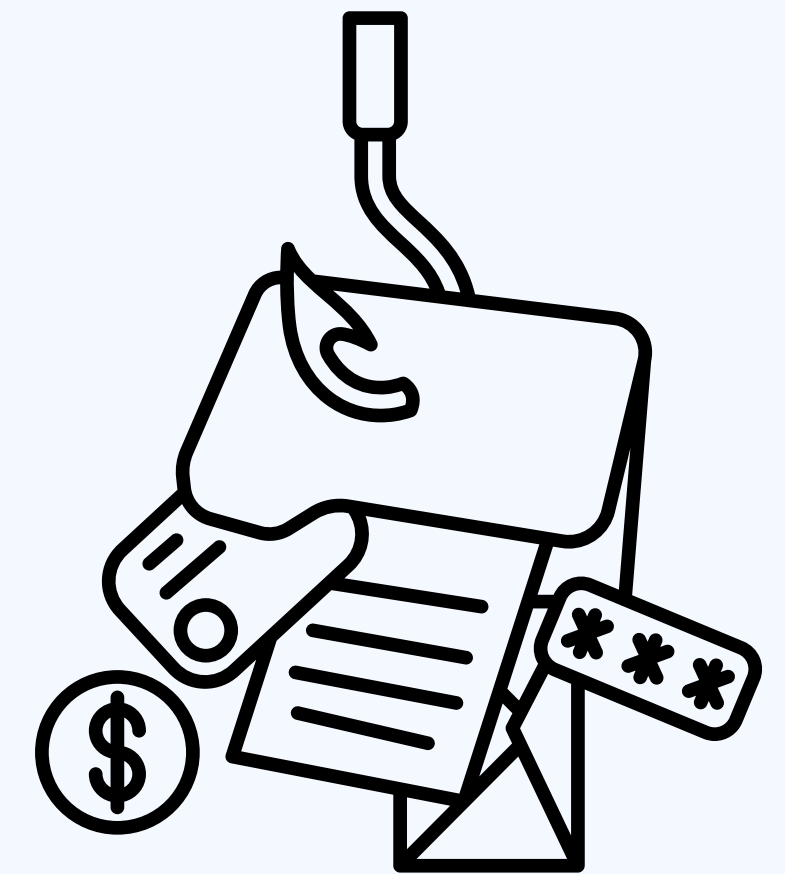
The initial breach was detected in December 2020.

3

They inserted malicious code into Orion updates, which were released to customers in March 2020.

4

The attack exploited a trusted software distribution channel.





ATTRIBUTION



The SolarWinds cyber attack, discovered in December 2020, is attributed to APT29, also known as Cozy Bear, a Russian state-sponsored group tied to the SVR. Experts, including the U.S. government (CISA, FBI, NSA), UK's NCSC, and private firms like Microsoft and FireEye, linked it to APT29 through the Sunburst malware, unique command-and-control patterns, and tactics matching prior espionage campaigns. Targeting U.S. agencies and NATO allies, the attack aligned with Russia's strategic goals amid geopolitical tensions. Evidence from compromised Orion software updates and intelligence reports led to U.S. sanctions on Russia in 2021, highlighting the attack's severity.

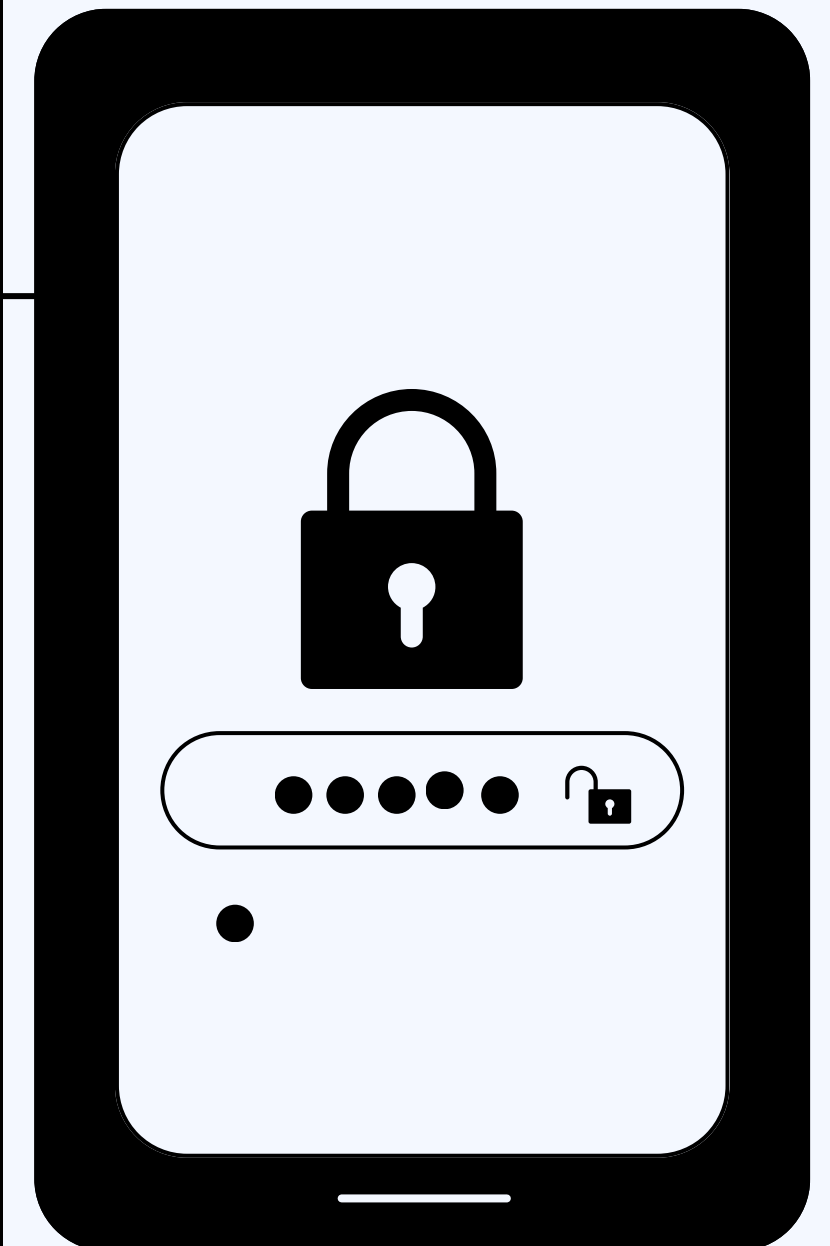
IMPACT

In IronNet's 2021 Cybersecurity Impact Report, 85% of SolarWinds cyber attack victims said the SolarWinds attack had an impact ranging from "small" to "significant."

One of the most notable impacts was the financial fallout from the attack. On average, the attack cost companies 11% of their annual revenue. The impact was more dramatic in the United States than across seas. In the U.S., the average impact was 14% of annual revenue, while U.K. companies suffered 8.6% losses. Those in Singapore had to deal with a financial hit of 9.1% of their annual revenue.

There were some positive impacts as well, with 81% of survey respondents saying they're more likely to share best practices with other organizations as a result of lessons learned during the attack.

Some had already started changing their practices. Sixty-seven percent had begun sharing information with their peers in the tech sector. Also, half had started sharing additional information with government agencies.



LESSION LEARNED

The SolarWinds attack revealed the dangers of supply chain vulnerabilities, stressing the need for better third-party oversight, stronger threat detection, and zero-trust security, as attackers exploited a trusted vendor to breach numerous targets undetected for months.





FUTURE CYBER THREATS

AI Advancements: Artificial intelligence is supercharging cybercrime by enabling highly convincing phishing emails, voice deepfakes, and malware that learns to dodge security tools in real time, making attacks harder to detect and stop.

Supply Chain Vulnerabilities: Attackers target weak links in vendor networks—like software providers or contractors—knowing a single breach, such as SolarWinds, can cascade to thousands of organizations, amplifying damage.

Geopolitical Tensions: Rising conflicts, like U.S.-China or Russia-West rivalries, drive state-backed hackers to hit critical sectors (e.g., energy, defense) for espionage, sabotage, or leverage in global power struggles

Quantum Computing: As quantum technology nears practical use, it could crack current encryption methods (like RSA), exposing sensitive data unless organizations shift to quantum-resistant defenses soon.



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THANKYOU

Stay Safe, Stay Secure



UNLOCKED ✓